

Amit Roy

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Education

National Institute of Technology, Silchar

Bachelor of Technology - Electronics and Communication Engineering | CGPA : 8.35

Assam, India

2025

Work Experience

Software Engineer @KHATA BUDDY

Apr 2026 - Present

- Built **Katlas**, an **autonomous AI software engineer** (Claude Opus + MCP) that takes a support ticket from **branch** → **implement** → **QA** → **human review** → **merge** → **production deploy** — with a crash-safe job queue, a fresh EC2 box per ticket, resumable work across review rounds, and a **confidence gate that hands risky changes back to a human**; **100+ production tickets shipped end-to-end** in ~2 months.
- Built a production **tool-using LLM agent** — one engine serving **web, mobile & WhatsApp** — that **executes real operations, not just answers**: **RAG on MongoDB Atlas**, **MCP tool registry**, **vision bill-scan**, **OpenAI** → **Gemini failover**, **LLM-judged relevancy scoring** + token/cost accounting so quality is measured, not assumed; drove AI to **~15%+ of core workflows**, cut manual data entry **~80%**.
- Profiling-driven performance engineering** across DB, service and client: fix cut a render path **827 ms** → **~100 ms**, document generation **~1 s** → **~550 ms**, interaction latency **~950 ms** → **40–90 ms**.
- Built the team's **delivery platform** — a **notification backbone** where a new alert stream is one routing entry (merge / deploy / Katlas / business events), a **GitHub Actions** → **S3** frontend pipeline, and a lock-guarded deploy runner.
- Extended the existing **billing platform** with prorated upgrades, coupons, custom plans, wallet and shipped a **self-serve admin console** for sales/ops — **~70%+ of billing operations now need no engineer**.

Software Development Engineer @ZUPEE

Jan 2025 - Feb 2026

- Built a **centralized, idempotent bootstrap** for **AWS SQS/SNS provisioning** across the org's **microservices** — queues and topics declared once, created **once at service startup** (safe to re-run, no drift) — retiring scattered per-service creation.
- Developed **payment & subscription workflows** processing **~30–40K transactions/day** — **Juspay / PhonePe** integrations, **payment-due notifications**, **retry jobs for failed deposits** — improving payment success and renewal rates.
- Developed the **user-segmentation system** — **static, live & derived segments** via query rules and daily pipelines — powering **segment-driven UI**, benefits, VIP handling and personalized offers.

Projects

AGENT DECK [GitHub](#) — control deck for mixed human + AI teams

Jul 2026 - Present

- Built a platform where a **team of AI agents and humans works in one channel** — and the **manager can itself be an agent**, owning and controlling the agents under it through an **ownership-derived permission engine** — no stored roles, one flat **Member** model, and any member can talk to any other (human-to-agent, agent-to-agent).
- Each agent is a **LangGraph**-orchestrated **ReAct** agent with its own provider / model / reasoning effort (**Claude / Gemini**), persistent memory and **auto-compacting context**; **Python + FastAPI**, **Repository**-protocol DI over **MongoDB Atlas**, **SSE** on **GCP Cloud Run**, **Google OIDC**; **React 19** frontend.

AGENT HIVE [GitHub](#) [LinkedIn Post](#) — live monitor + shared channel for AI coding agents

Jun 2026

- Built a browser **control room for AI coding agents** on the **Claude Agent SDK** — every session tailed **live**, agents launched / resumed / steered from the UI, and a **permission gate**: safe tools auto-run, risky ones (**Bash / Write**) pause for an **Allow / Deny** card over **SSE**.
- Designed a **shared team channel** where agents and humans talk in one thread — per-agent **MCP tools**, exactly-once delivery, loop guard against agent-to-agent ping-pong; **Claude**, **Codex**, **Antigravity** and a **human dev** debated trade-offs in-channel and converged on a pinned decision.

AI NOTES + AI TYPING PLATFORM [npm](#) [GitHub](#)

Jul 2026 - Present

- AI Notes (Live)** — rich-text notes with an **AI edit box**: tell it what to change and **Gemini rewrites only that** (schema-constrained JSON, editor-safe HTML) or answers a question about the note in place; **anonymous-first**, data claimed on Google sign-in.
- AI Typing Practice (Live)** — describe what you want to type (code, interview problems, any subject, even in Hinglish) and **Gemini generates the passage** in the right format (code / Q→Reasoning→Answer / prose), with live **WPM + accuracy**.
- Stack**: **React 19** + **TypeScript** + **Tailwind v4**, **Express** + **MongoDB**, **Gemini** behind a port; **@roshx/core** + **@roshx/ui** published as **open-source npm packages** (one contract for server and web, breaks at compile time); **Docker** → **GCP Cloud Run** via **GitHub Actions**.

Achievements/Awards

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|---------------------|---|-------------------------|
| • LeetCode | Guardian Rating 2181 (Top 1.01%) 1800+ solved | Profile |
| | Rank 169/35040 (A.I.R 19) in Weekly 412 Rank 203/25287 (A.I.R 34) in Biweekly 129 | |
| • Codeforces | Expert Rating 1615 700+ solved CodeChef 3-star GFG 4-star | Profile |

Skills

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|-------------------------------|--|
| • Languages & Core | C/C++ , JavaScript , Python , TypeScript Data Structures & Algorithms , OOP |
| • LLM / GenAI | Tool-using Agents , RAG , MCP , LangGraph , Evaluation , Multimodal , Claude / OpenAI / Gemini |
| • Backend & Web | Node.js , Express.js , FastAPI , Kafka , MongoDB (Atlas) , Redis React.js , Tailwind CSS |
| • Cloud & DevOps | AWS (EC2, S3, SQS/SNS) , GCP Cloud Run , Docker , Kubernetes , CI/CD , Git , OAuth/OIDC |